

MEU Transport System

Smart Fleet Management & Live Tracking

Prepared for:
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1 Executive Summary

The MEU Transport System is an integrated IoT-style solution designed for the Middle East University. It solves the critical problem of real-time student commute tracking by bridging a **React-based Admin Dashboard** with a **React Native Mobile Application** via a high-frequency **Firebase Firestore** backend.

2 System Architecture

The architecture follows a tri-node synchronization pattern:

1. **Data Ingestion:** Driver App sends GPS and occupancy data every 2s.
2. **Data Processing:** Firebase Firestore handles sub-second real-time state updates.
3. **Data Visualization:** Admin Dashboard renders live bus markers and route polylines.

3 Data Modeling

Below are the formal schemas for the primary data collections.

Bus State Model

$$\text{Bus}_{id} = \begin{cases} \text{Location} & (lat, lng) \\ \text{Load} & \text{passengers} \in \mathbb{N} \\ \text{Status} & \{\text{Active}, \text{Idle}, \text{Offline}\} \end{cases}$$

Route Logistics Model

All routes are mathematically forced to converge at the central node:

$$\forall \text{Route}_i, \quad \text{Destination} = (31.81^\circ\text{N}, 35.92^\circ\text{E})$$

4 Technical Implementation Details

4.1 Mobile Navigation & Side Menus

To improve UX, we implemented a **Drawer Navigator** pattern. This was integrated using the following library stack:

- `@react-navigation/drawer` for the UI container.
- `react-native-reanimated` for fluid menu transitions.

4.2 GPS Accuracy Overrides

For iOS-specific deployment, the following configuration was applied to ensure the device remains in "High Performance" GPS mode:

- **Accuracy Level:** `Location.Accuracy.BestForNavigation`
- **Activity Type:** `Fitness/Automotive`

5 Security & Authentication

Authentication is managed via **Firestore Auth** with persistence stored locally on the device using **AsyncStorage**. Administrative access is restricted via Firestore Rules checking for `auth.token.role == 'admin'`.

6 Maintenance & Error Pipelines

Drivers can report system issues directly to the admin. These reports are stored as follows:

$$\text{Report} = \{\text{BusID}, \text{IssueText}, \text{DriverEmail}, \text{Timestamp}\}$$

Admins can view and resolve these from the **Fleet Personnel** section of the dashboard.